MACHINE LEARNING FUNDAMENTALS

ShopSmart Retail – ML01 Foundations Report

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1. Introduction

This report is prepared for ShopSmart Retail Ltd. As part of the AnalystLab Africa Experience Lab, Machine Learning Track, Week 1. ShopSmart is a retail chain operating across Nigeria and wants to begin using artificial Intelligence to improve its operations. The company has hired AnalystLab Africa to build its first machine learning prototypes: one to predict the value of customer orders, and another to predict whether an order will be returned. This report explains what machine learning is, the main types of machine learning, the difference between regression and classification, and the workflow that will be followed to build these models.

2. What is Machine Learning?

Machine learning is a way of teaching computers to make decisions and predictions from data, without a human having to write every rule by hand. Instead of programming exact instructions, we give the computer examples and let it discover patterns on its own.

In traditional programming, a developer writes rules, feeds in data, and the computer produces answer. In machine learning, the direction is reversed: we give the computer data and the correct answers, and the computer produces the rules (called a "model"). Those rules can then be applied to new, unseen data.

For example, at ShopSmart, to predict whether an order will be returned, we do not write thousands of "if-then" rules. Instead, we feed the computer 50,000 past orders where we already know whether each was returned. The computer learns the patterns that lead to a return, for example, certain products, purchase times, or customer profiles, and can then estimate the likelihood of a return for a new order.

3. Type of Machine Learning

3.1. Supervised Learning

Supervised learning is machine learning with labelled data. Every training example has an input and known correct answer (label). The model learns to map inputs to labels, so it can predict the label for new data it has never seen.

ShopSmart examples:

- Predicting the total value of an order (a number)
- Predcting whether an order will be returned (yes or no)

Common algorithms:

Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Support Vector Machines.

3.2. Unspervised Learning

Unsupervised learning works with data that has no labels. The model is not told the "right answer" instead, it finds structure, groupings, or hidden patterns by itself.

ShopSmart examples:

- Grouping customers into segments such as "budget shoppers", "loyal premium customers", and "seasonal buyers" for targeted marketing.
- Detecting unusual transactions that might indicate fraud.

Common algorithms:

K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA), Isolation Forest.

3.3. Reinforcement Learning

Reinforcement learning is a type of machine learning where an agent learns by interacting with an environment. The agent takes actions and receives rewards or penalties, and over time it learns a policy that maximises its total reward. This is similar to training a pet: good behaviour is rewarded and bad behaviour is discouraged.

ShopSmart examples:

- A dynamic pricing agent that adjusts prices to maximise profit.
- An inventory agent that learns the best time to reorder stock to avoid shortages or excess.

Key concepts:

Agent, environment, state, action, reward, policy, exploration vs exploitation.

4. Regression vs classification

Regression and classification are the two main types of supervised learning. Regression predicts a number, while classification predicts a category.

	Regression	Classification
What it predicts	A number	A category
Question answered	“How much?”	“Which one?”
ShopSmart Example	Order value in naira	Returned: Yes or No
Example algorithms	Linear Regression, Random Forest Regressor	Logistic Regression, Decision Tree Classifier
Evaluation metrics	MAE, RMSE, R^2	Accuracy, Precision, Recall, F1

Note: Despite its name, Logistic Regression is a classification algorithm. The word "regression" in its name refers to the type of mathematical function it uses, not the type of problem it solves.

Regression and classification are evaluated differently. For regression we ask how far our numeric predictions are from the true values, using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. For classification we ask how often we predict the right class, using metrics such as Accuracy, Precision, Recall, and F1-Score. Precision and Recall matter especially when the classes are imbalanced for example, when only 5% of orders are returned.

5. The Machine Learning Workflow

A machine learning project follows a structured workflow. It is not a straight line if a model performs poorly, we return to earlier stages and try again.

1. **Business Understanding** — Define the business problem and the decision the model will support. At ShopSmart: reduce product returns and forecast order value.
2. **Data Collection** — Gather relevant data. For ShopSmart this includes the ten datasets provided: Customers, Products, Orders, Stores, Returns, Employees, Suppliers, Inventory, Order Items, and the Data Dictionary.
3. **Data Preparation** — Clean the data, handle missing values, remove duplicates, merge tables, encode categorical fields, and split the data into training and testing sets. This stage typically consumes 60–80% of the total project time.
4. **Exploratory Data Analysis and Feature Engineering** — Explore distributions, look for correlations, and create new features such as order month, discount percentage, and customer age group.
5. **Model Training** — Train baseline models first. For ShopSmart: Linear Regression to predict order value and Logistic Regression to predict returns. More complex models are only justified if they beat the baseline.
6. **Model Evaluation** — Assess performance using appropriate metrics. Regression: MAE, RMSE, R^2. Classification: Accuracy, Precision, Recall, F1, Confusion Matrix.
7. **Deployment** — Save the trained model to disk (e.g., using Joblib) and integrate it into a system that other people can use.
8. **Monitoring and Retraining** — Monitor the model in production for data drift (input distribution changes) and concept drift (the relationship between inputs and outputs changes). Retrain the model when performance degrades.

6. Conclusion

ShopSmart has rich historical data and clear business problems: unpredictable sales and a rising rate of product returns. Machine learning offers a structured approach to turn that data into actionable decisions. This report has defined the core concepts: what machine learning is, the three main types, the difference between regression and classification, and the workflow that a typical project follows. These foundations will guide the rest of Week 1, where we will prepare the ShopSmart dataset and build two baseline models: predicting order value and predicting product returns.

References

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